

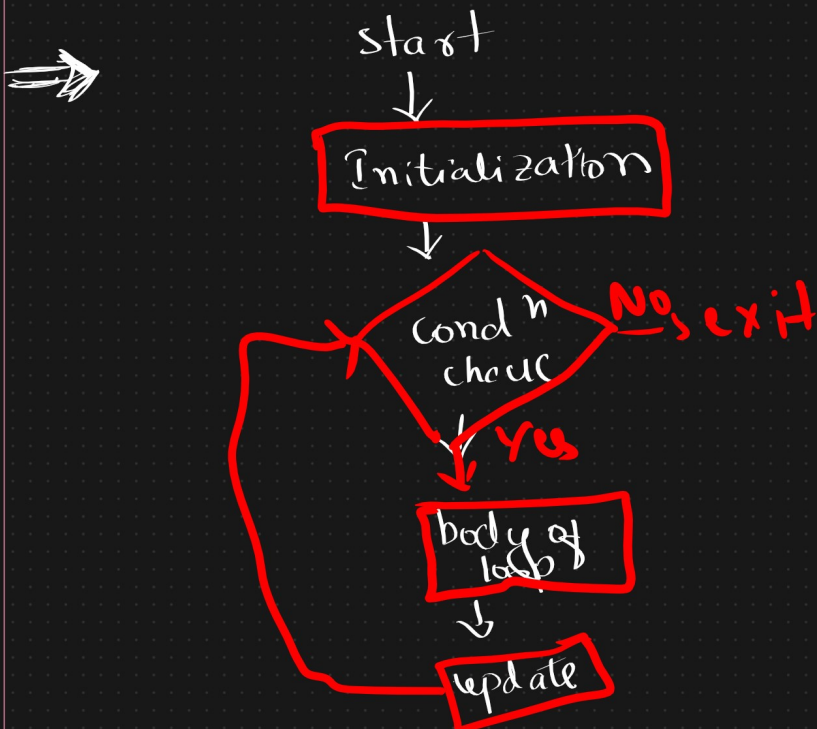
⇒ 10 hours == Focus 1 hour

⇒ variable
Datatype
Type casting
operators
conditional } loops
pattern programming

⇒ loop ⇒ for loop
while
do-while }

S. println();
[**]

* S.o.p(*); * loops
* S.o.p('A');
*
*
*
(



*
 *
 *
 *
 *
 *

for (initialization; condition; update)
 {
 }

Body of loop

for (int i = 0; i < 5; i++)
 {
 }

s.o. println(" ");

i 0 1 2 3 4 5
 0 < 5 -> T
 1 < 5 -> T
 2 < 5
 3 < 5
 4 < 5
 5 < 5

i 0 1 2 3 4 5
 0 < 5
 1 < 5
 2 < 5
 3 < 5
 4 < 5
 5 < 5

*
 *
 *
 *
 *

=y

+++++

{
 }

for (int i = 0; i < 5; i++)
 {
 }

s.o. println(" ");

i 0 1 2 3 4 5
 0 < 5
 1 < 5
 2 < 5
 3 < 5
 4 < 5
 5 < 5

i 0 1 2 3 4 5
 0 < 5
 1 < 5
 2 < 5
 3 < 5
 4 < 5
 5 < 5

while loop :-

int i = 0;

while (i < 5) {

{
 }

s.o. println(" ");
 i++;

i 0 1 2 3 4 5

i < 5

0 < 5

1 < 5

2 < 5

3 < 5

4 < 5

*
 *
 *
 *
 *

do-while:-

i 0 1 2 3 4 5

int i = 0;

do

{ s.o.p (" ");

i++;

} while (i < 5);

1 < 5

*

*

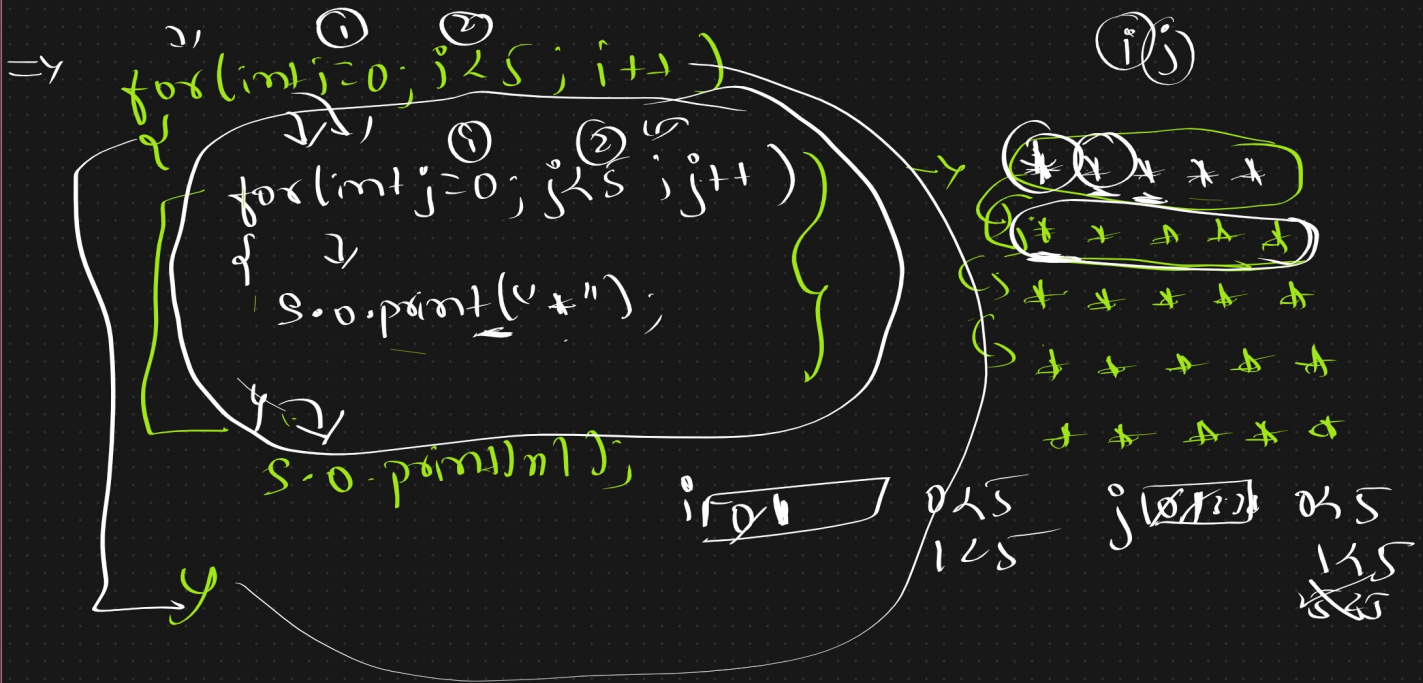
=> same task -> repetitive -> loops => ✓

for -> ✓

while -> ✓

do-while -> ✓

{ for-each
(enhanced loop)
=> array



{ if -> else } => ✓

Operator & logical operators

|| && !

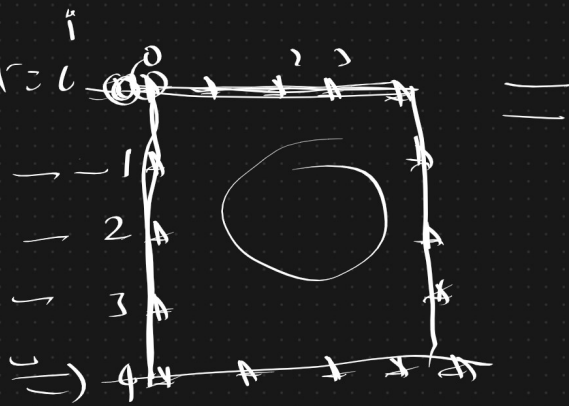
{ i==0 || j==0 || }
i!=0 || j!=0

if ($i == 0 || i == 4 || j == 0 || j == 4$) $r = 1$

so print (4 * " ");

else

so print (" ");



$$n = 5$$

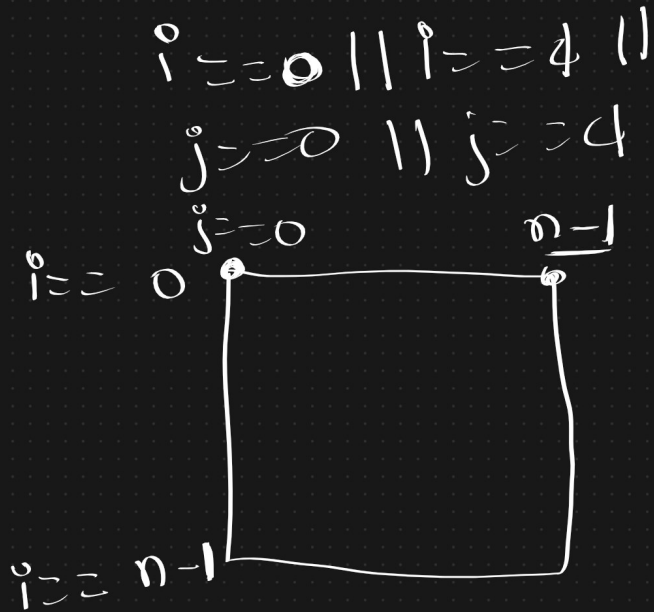
$$n - 1 = 4$$

$$4 (0 \text{ to } n - 1)$$

$$5 - 1$$

$$\frac{n}{2}$$

$$4$$



$$i == 0 || i == n - 1 || j == 0$$

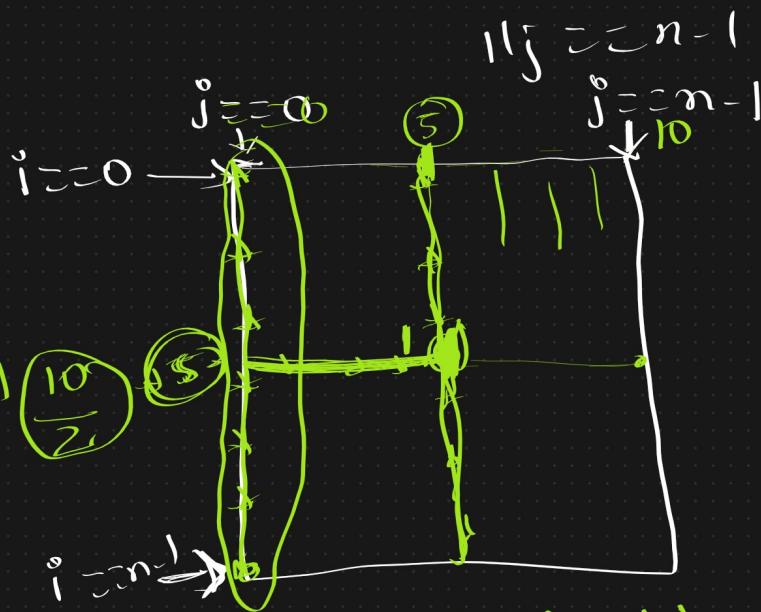
$$n - 1 = 10$$

$$n = 11$$

$$n - 1 = 10$$

$$\frac{n - 1}{2}$$

$$\frac{11 - 1}{2} = 5$$



$$j == 0 || j == \frac{n - 1}{2} ||$$

$$i == \frac{n - 1}{2} \&\& j \neq n - 1$$

$$\frac{n}{2} \&\&$$

$j=0$ & $i=0$ & $i=n-1$ $i=0$
 (00)

$j=0$ & $i>0$ & $i \leq n-1$ $i=\frac{n-1}{2}$

$j=0$ & $i>0$ & $i \leq n-1$ $i=n-1$
 || $i=0$ & $j>0$ & $j \leq n-1$ ||

$i=n-1$ & $j>0$ & $j \leq n-1$

$(i=0 \text{ & } j>0 \text{ & } j \leq n-1)$ ||

$i=n-1 \text{ & } j>0 \text{ & } j \leq n-1$ ||

$j=0 \text{ & } i>0 \text{ & } i \leq n-1$ $\frac{n-1}{2}$

$j=n-1 \text{ & } i>0 \text{ & } i \leq n-1$

||

$i=0$

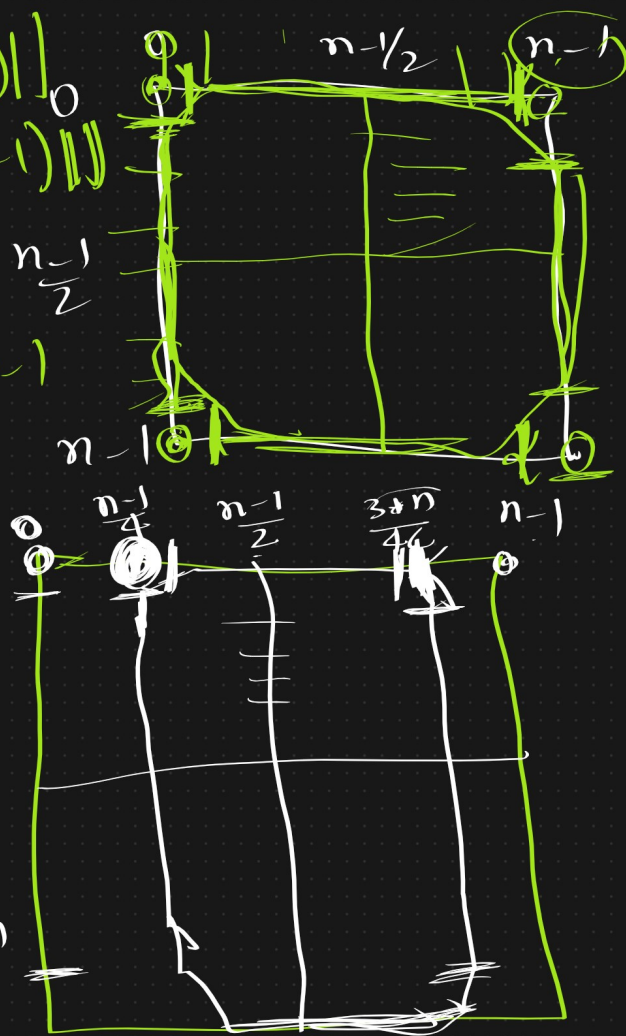
$\frac{n-1}{4}$ $\frac{10}{4} \Rightarrow 2.5$

$\frac{3 \times n}{4}$ $\frac{30}{4} \Rightarrow 7.5$ $\frac{n-1}{2}$

$i=0$ & $j \leq \frac{n-1}{4}$ & $j \leq \frac{3 \times n}{4}$
 $n-1$

$j=\frac{n-1}{4}$ & $i>0$ & $i \leq n-1$ ||

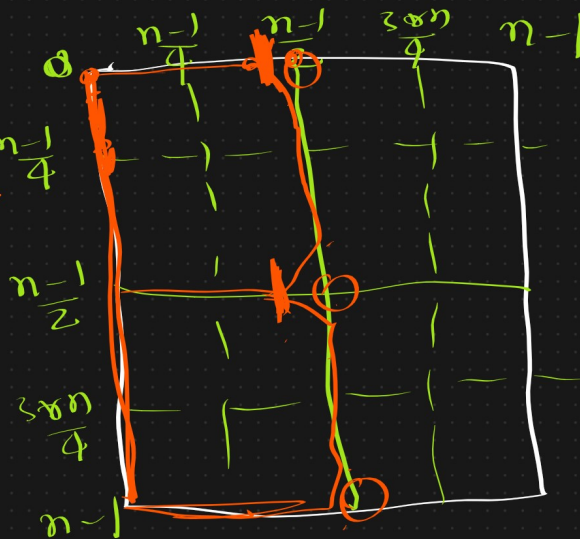
$j=\frac{3 \times n}{4}$ & $i>0$ & $i \leq n-1$



$\text{for}(i)$
 $\text{for}(j)$
 $\text{sum} = \text{sum} + (a[i][j] + 1)$
 $\text{sum} = \text{sum} + \text{len}()$

$\text{for}(i)$
 $\text{for}(j)$
 $(\text{sum} + 1) - 1$
 $(\text{sum} + 1) - 1$
 $\text{sum} = \text{sum} + \text{len}()$

$j = 0 \parallel$
 $j = n-1 \parallel$
 $i = 0 \parallel j = n-1 \parallel$
 $i = n-1 \parallel j = 0 \parallel$



$i = n-1 \parallel j = n-1 \parallel$
 $i = n-1 \parallel j = n/2 \parallel$

A B C D E F G H I J (K) L (M N)
 O P (Q) (R) S T (U V W) (X Y Z)

$\Rightarrow$  variables $\Rightarrow$ ✓
Datatypes $\Rightarrow$ ✓
Typelasting $\Rightarrow$ ✓
if-else ladder $\Rightarrow$ ✓
operators $\Rightarrow$ ✓
loops $\Rightarrow$ ✓



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